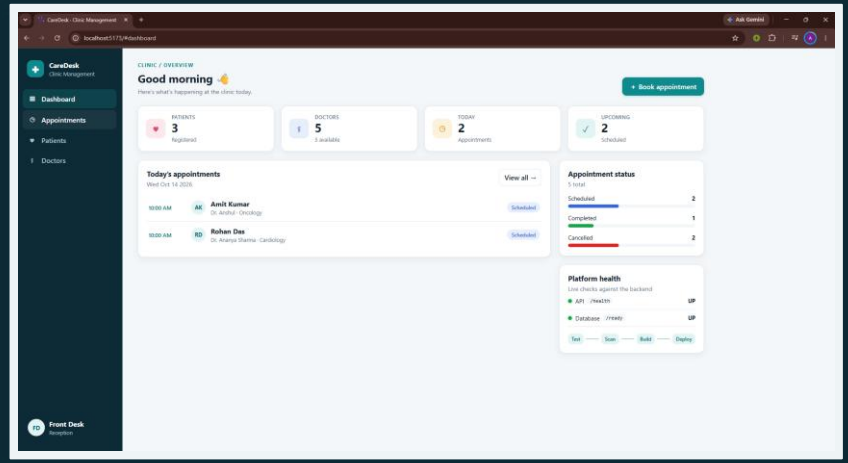
DEVSECOPS CAPSTONE · SESSION 21

CareDesk

A clinic management app, delivered end to end by a full DevSecOps pipeline



github.com/aryen1101/clinic_managment_devops

The problem CareDesk solves

Small clinics today

Paper registers
Patient details and visits are hard to find

Double-booked doctors
Two patients get the same slot

No overview
Nobody sees today's schedule at a glance

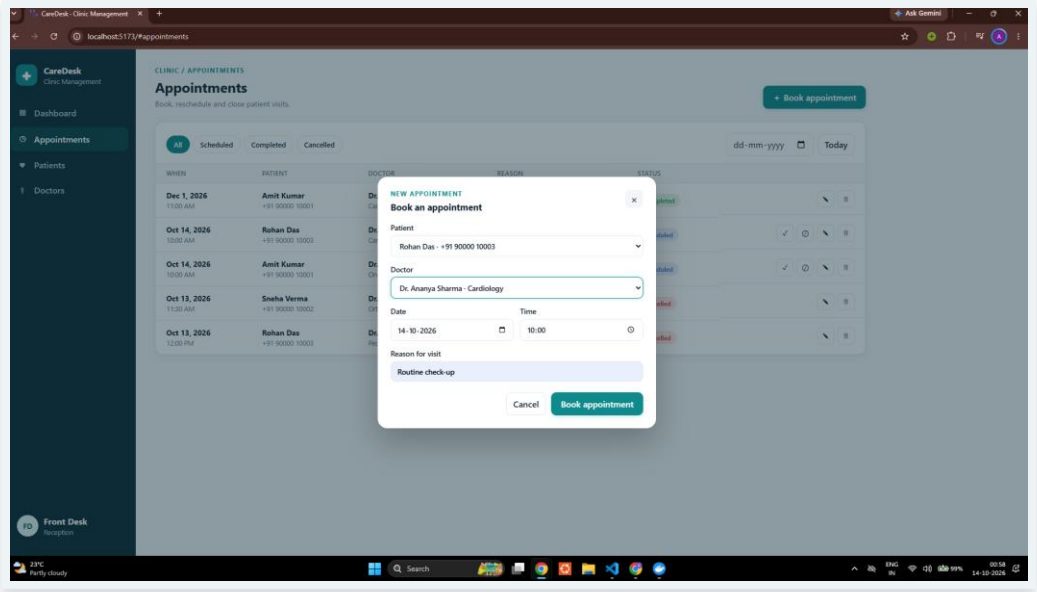
With CareDesk

Patient registry
Search, profiles and full visit history

Safe booking
A doctor can never be double-booked (HTTP 409)

Live dashboard
Today's appointments, status and system health

What I built



3

linked tables: patients, doctors, appointments

20+

REST endpoints with Swagger docs

24

automated API tests (pytest)

Technology stack

FRONTEND
React 19 + Vite

BACKEND
FastAPI · Python 3.13

DATABASE
PostgreSQL + Alembic

TESTING
Pytest · 24 tests

CONTAINERS
Docker + Compose

CI/CD
GitHub Actions + GHCR

SECURITY
Trivy · Bandit · Gitleaks

INFRASTRUCTURE
Terraform · AWS EKS

ORCHESTRATION
Kubernetes

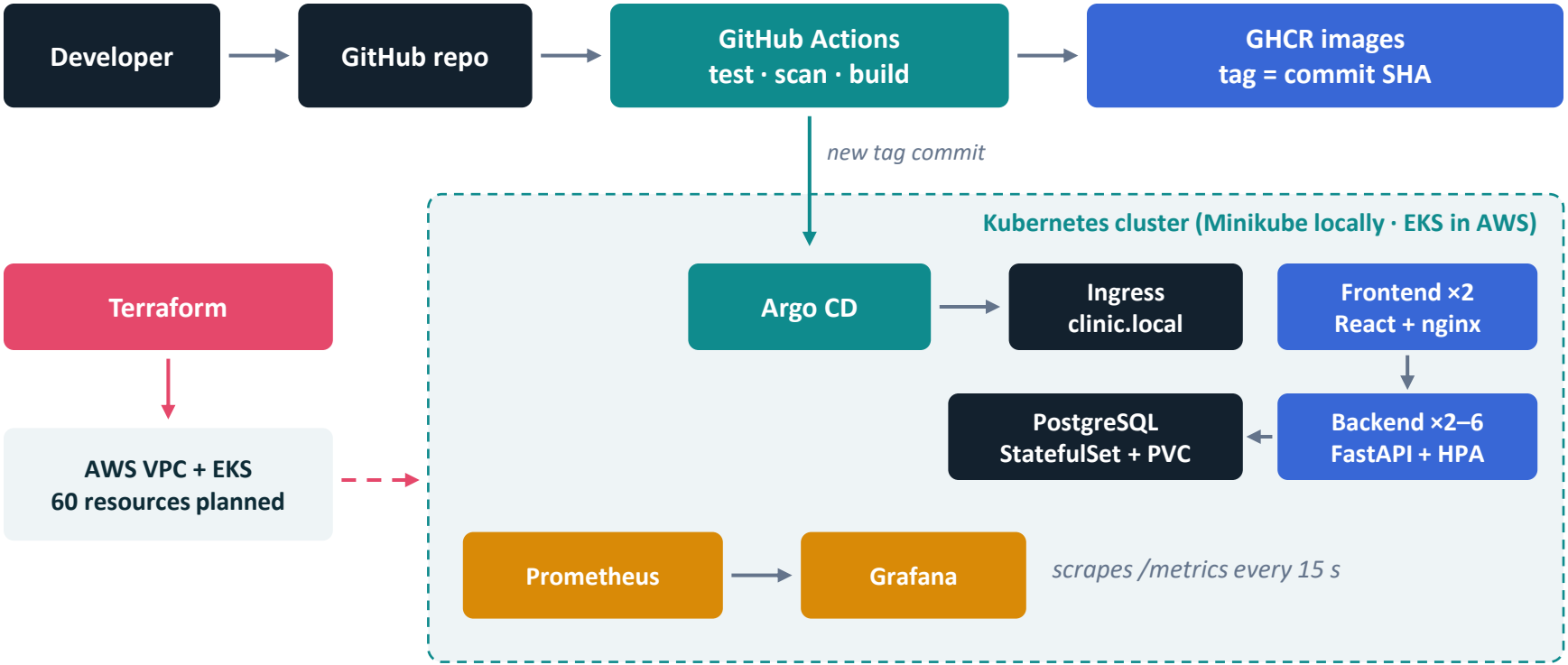
PACKAGING
Helm 3

GITOPS
Argo CD

OBSERVABILITY
Prometheus + Grafana

Why Kubernetes: self-healing, rolling updates, autoscaling · alternatives considered: Docker Swarm, AWS ECS

Architecture overview



CI/CD pipeline on every push to main

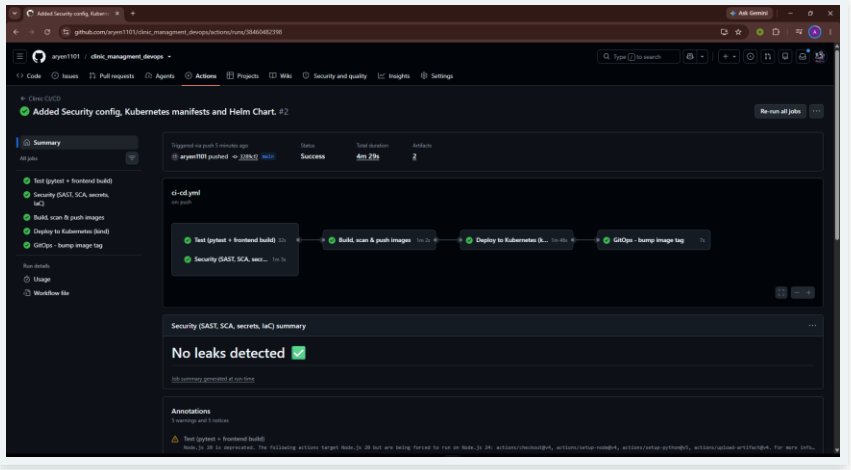
1 Test
ruff + 24 pytest
frontend build

2 Security
Bandit · pip-audit
npm audit · Gitleaks

3 Build & scan
Docker build
Trivy gate

4 Push
GHCR, tag =
commit SHA

5 Deploy
kind smoke test
GitOps tag bump



Trigger
Every push and pull request to main

Traceable images
Tag = Git commit SHA, never latest

Quality gate
A failing test stops the pipeline: nothing is pushed

Security at every layer

SAST · Bandit
Insecure patterns in my Python code

SCA · pip-audit + npm audit
Known CVEs in dependencies

Secrets · Gitleaks
Passwords or keys in Git history

IaC · Trivy config
Misconfigured Dockerfiles, K8s, Terraform

Containers · Trivy image
Fails the build on HIGH or CRITICAL CVEs

47 → 0

HIGH CVEs after hardening the images: apk upgrade in the nginx image, pip removed from the Python image



Containers and Kubernetes

Non-root images
Backend uid 10001 · frontend nginx uid 101

Multi-stage build
Node builds it, nginx serves it

Health checks + probes
/health liveness · /ready checks the database

ConfigMap + Secret
Config and DB password outside the image

```
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops> kubectl get pods,svc,ingress,hpa,pvc -n clinic
NAME                                 READY   STATUS    RESTARTS   AGE
pod/clinic-backend-79b8f774c-cqg58   1/1     Running   0          61s
pod/clinic-backend-79b8f774c-v7lfl   1/1     Running   0          61s
pod/clinic-frontend-78ff44bc8f-d4pnp 1/1     Running   0          61s
pod/clinic-frontend-78ff44bc8f-hwjx8 1/1     Running   0          61s
pod/clinic-postgres-0                1/1     Running   0          61s

NAME                       TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
service/clinic-backend     ClusterIP   10.97.136.42     <none>        8000/TCP   61s
service/clinic-frontend    ClusterIP   10.96.26.253     <none>        80/TCP     61s
service/clinic-postgres    ClusterIP   10.110.121.220   <none>        5432/TCP   61s

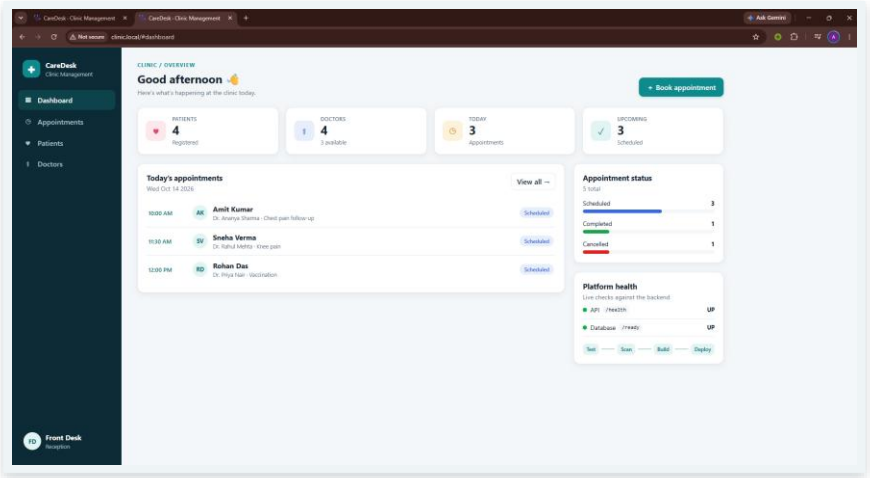
NAME                              CLASS   HOSTS          ADDRESS        PORTS   AGE
ingress.networking.k8s.io/clinic  nginx   clinic.local   192.168.49.2   80      61s

NAME                                                  REFERENCE                      TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/clinic-backend    Deployment/clinic-backend      cpu: <unknown>/70%   2         6         2          61s
horizontalpodautoscaler.autoscaling/clinic-frontend   Deployment/clinic-frontend     cpu: <unknown>/80%   2         4         2          61s

NAME                                        STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
persistentvolumeclaim/data-clinic-postgres-0   Bound    pvc-3007e5f7-e50e-4304-945b-b36e5aa6e948   1Gi        RWO            standard       <unset>                 61s
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops> helm list -n clinic
NAME    NAMESPACE       REVISION        UPDATED                                 STATUS          CHART           APP VERSION
clinic  clinic          1               2026-10-14 12:40:44.4566076 +0530 IST   deployed        clinic-1.0.0    1.0.0
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops>
```

Helm release: 2 backend + 2 frontend + PostgreSQL pods, ClusterIP services, Ingress, HPA and a bound PVC

Ingress and autoscaling under load



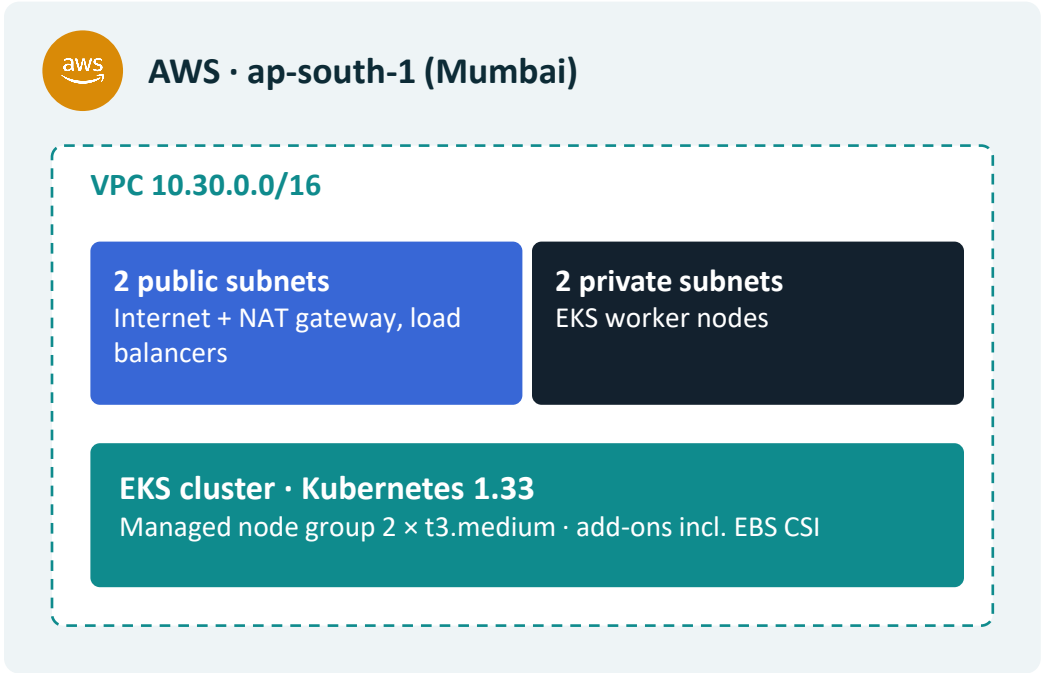
http://clinic.local through the nginx Ingress

```
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops> kubectl get hpa -n clinic -w
NAME             REFERENCE                   TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
clinic-backend   Deployment/clinic-backend   cpu: 2%/70%   2         6         2          36m
clinic-frontend  Deployment/clinic-frontend  cpu: 2%/80%   2         4         2          36m
clinic-backend   Deployment/clinic-backend   cpu: 279%/70%   2         6         2          38m
clinic-backend   Deployment/clinic-backend   cpu: 279%/70%   2         6         6          38m
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops> kubectl get hpa -n clinic
NAME             REFERENCE                   TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
clinic-backend   Deployment/clinic-backend   cpu: 279%/70%   2         6         6          38m
clinic-frontend  Deployment/clinic-frontend  cpu: 2%/80%   2         4         2          38m
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops> kubectl get pods -n clinic -l app=clinic-backend
NAME                              READY   STATUS    RESTARTS   AGE
clinic-backend-79b8f774c-2bm55    1/1     Running   0          62s
clinic-backend-79b8f774c-6f8t4    1/1     Running   0          62s
clinic-backend-79b8f774c-cqg58    1/1     Running   0          39m
clinic-backend-79b8f774c-jsbf8    1/1     Running   0          62s
clinic-backend-79b8f774c-jsjrv    1/1     Running   0          62s
clinic-backend-79b8f774c-v7lfl    1/1     Running   0          39m
PS C:\Users\aryen\OneDrive\Documents\Desktop\Learning\clinic_managment_devops>
```

2 → 6

backend pods when CPU reached 279% of the 70% target

Infrastructure as Code with Terraform



60

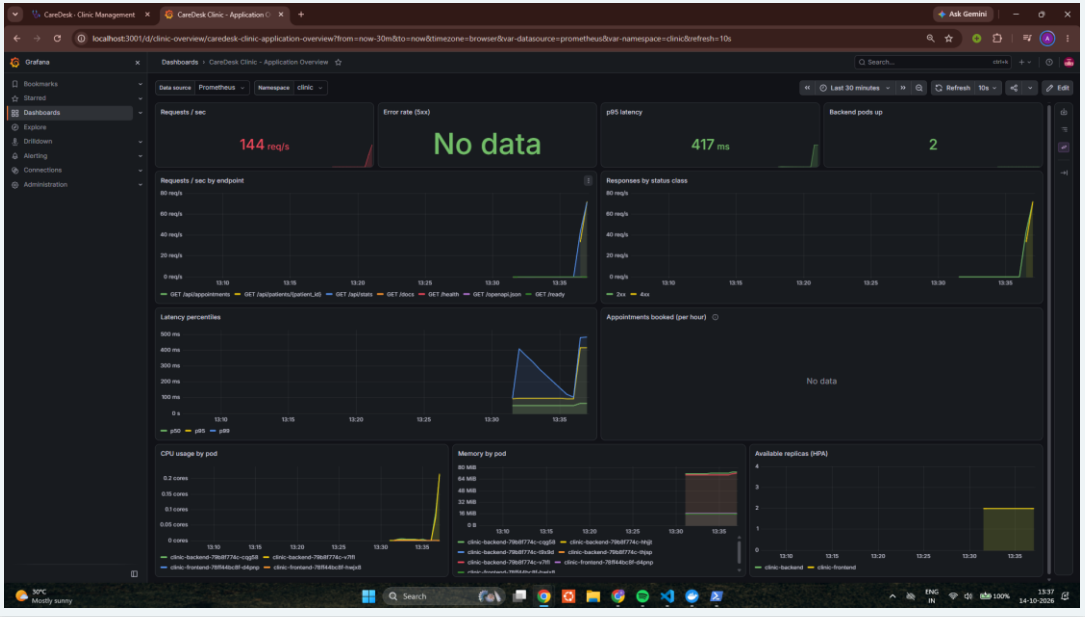
AWS resources in terraform plan

init · fmt · validate · plan
All run against a real AWS account

apply skipped on purpose
EKS and NAT are not in the free tier

No credentials in Git
tfvars and state are git-ignored

Monitoring and observability



Prometheus
Scrapes /metrics every 15 seconds

Grafana dashboard
11 panels: traffic, errors, latency, CPU

3 alert rules
Backend down · errors · slow p95

Business metric
Appointments booked over time

GitOps with Argo CD

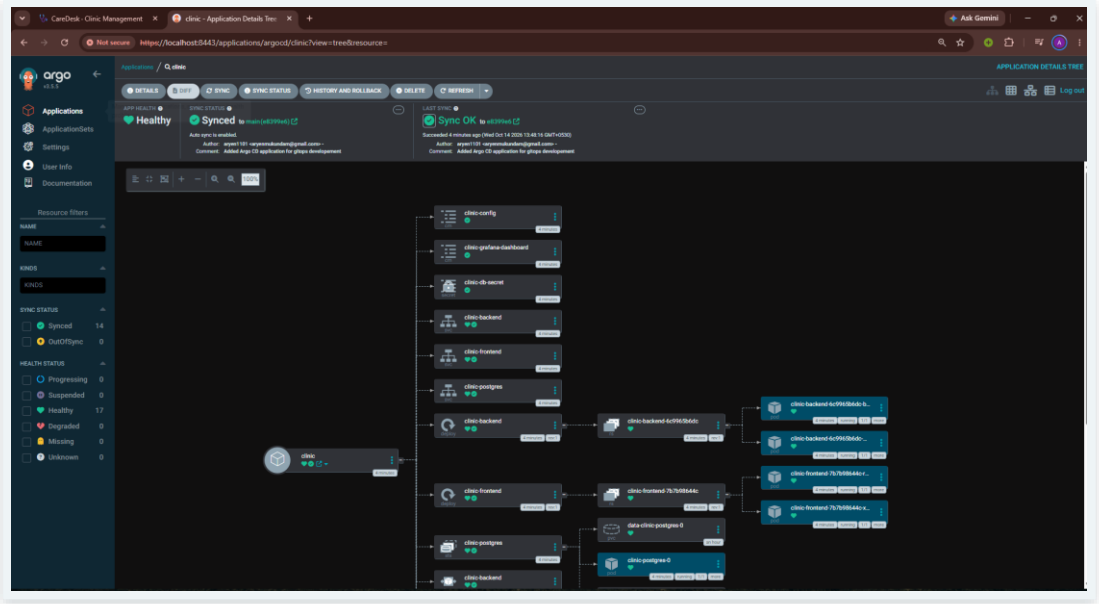
git push

CI builds image (SHA)

CI commits new tag

Argo CD syncs

Rolling update



Auto-sync with prune and self-heal: Git is the single source of truth, and manual kubectl changes are reverted

Quality gates that stop bad code

24

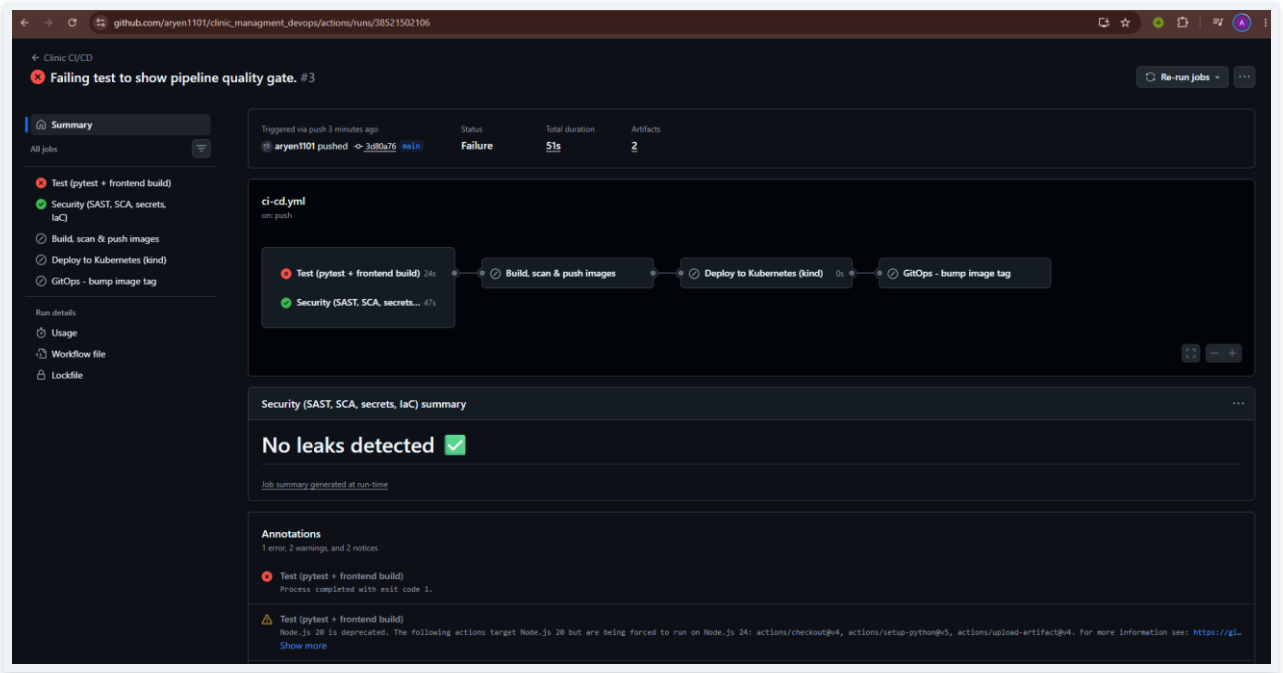
API tests on an isolated test database

5

security scanners in the pipeline

0

HIGH or CRITICAL CVEs in the images



A deliberately broken test turned the pipeline red: no image was built, pushed or deployed

Lessons learned

Readiness ≠ liveness
/health ignores the DB; /ready checks it

Startup order matters
Wait-for-DB + migration lock: zero restarts

Scanners find real issues
47 HIGH CVEs, fixed in the base images

Tag with the commit SHA
Every pod traces back to one commit

NEXT STEPS

Amazon RDS · TLS with cert-manager · staff login and roles · central logs with Loki

Thank you · github.com/aryen1101/clinic_managment_devops